

Patrick H. Jacobs
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Richmond, VA
804.564.3992

EDUCATION

University of Virginia, School of Engineering and Applied Science
B.S. Mechanical Engineering, Material Science Minor
GPA: 3.70

Charlottesville, VA
August 2023 - Present

WORK EXPERIENCE

BrownGreer PLC, Claims Reviewer

Summer 2024

- Worked on a team reviewing claims for the 3M Earplug lawsuit.

PROJECTS

Formula SAE Team Member

August 2024 - Present

- Participating in design tasks to create a car for competition at the Formula SAE Electric event in Michigan.
- Continuing development of CAD and CAM skills, while building a basic understanding of vehicle dynamics and mechanical systems.
- Tasked with designing wheel uprights, from material choice through to shape optimization using finite element analysis and accounting for manufacturability.

Electric Bike Personal Project

July 2025 - Present

- Sourced electrical components and assembling an electric bicycle on a stock hardtail frame, with the goal of learning more about integrating electrical and mechanical systems.

UVA In Germany: Global Technology Practices

Summer 2025

- Studied abroad in Germany, visiting several manufacturing and headquarter sites of companies including Mercedes, Bosch, and BASF while learning more about German culture and engineering.

LEADERSHIP, HONORS, AND EXTRACURRICULAR ACTIVITIES

First Robotics Competition Team 5804, Team Member / Captain

Spring 2019 –2023

- Collaborated with a team of 40+ people to construct a new robot each year to complete complex tasks in the annual First Robotics Competition.
- Competed and won several qualifier and district events. Qualified for the international competition and placed within the top 30 of 3,000+ teams in both the 2022 and 2023 seasons.
- Developed leadership and technical skills while dedicating 1,000+ hours of work over the course of 5 seasons.
- Developed mechanical and CAD modeling skills; one of two people to self-teach basic FEA, CAM skills and usage of a small CNC mill, modeled and milled a large number of the team's custom parts. The team received a Creativity Award at the international competition for our robot's design.

Leukemia and Lymphoma Society, Student of the Year Candidate

Spring 2021

- Led a team of 12 individuals to fundraise for children's blood cancer research.
- Organized and held meetings with local businesses to conduct events. Raised roughly \$47,000 for blood cancer research.

GENERAL SKILLS

Technical: Solidworks CSWA, Autodesk Inventor, Fusion 360, Python, Matlab, Microsoft Excel, Microsoft Word
Design: Computer Aided Design and Manufacturing, 3 and 4 Axis machining, basic finite element analysis